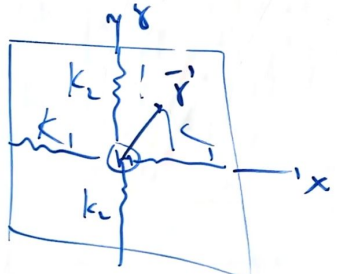
27.2.26

Wave Lec 21?

Coupled oscillations with two degrees of freedom



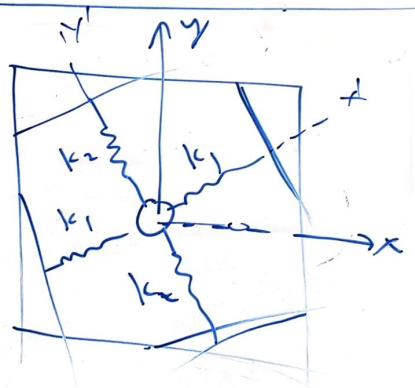
$$\ddot{x} \approx -\frac{2k_1}{m}x \Rightarrow \omega_1^2 = \frac{2k_1}{m}$$

$$\ddot{y} \approx -\frac{2k_1}{m}y \Rightarrow \omega_2^2 = \frac{2k_2}{m}$$

$$x = A_1\cos(\omega_1 t + \varphi_1)$$

$$y = A_2\cos(\omega_2 t + \varphi_2)$$

$$\vec{r} = \hat{i}\, A_1\cos(\omega_1 t + \varphi_1) + \hat{j}\cos(\omega_2 t + \varphi_2)$$



$$\ddot{x} = -a_{11}x - a_{12}y$$

$$\ddot{y} = -a_{21}x - a_{22}y$$

$$a_{12} = a_{21}$$

Anormal mode:

Every degree of freedom motion
- oscilate with same freq.
- ', ', same phase
- equil. at the same time

$$x = A\cos(\omega t + \varphi)$$

$$y = B\cos(\omega t + \varphi)$$

(1) (2) (3) (4) (5) (6) (7) (8) (7)

motion

e

hese

ame time

(1) $\ddot{x} = -4x - 2y$

(1) $\ddot{y} = -2x - 4y$

(3) $x = A\cos(\omega t+\varphi)$, $y = B\cos(\omega t+\varphi)$

(4) $\ddot{x} = -\omega^2 x$, $\ddot{y} = -\omega^2 y$

(5) $-\omega^2 x = -4x - 2y$

(6) $-\omega^2 y = -2x - 4y$

(7) $(4-\omega^2)x + 2y = 0$ a)

(8) $2x + (4-\omega^2)y = 0$

(7) $\Rightarrow \frac{y}{x} = \frac{\omega^2-4}{2}$,

(8) $\Rightarrow \frac{y}{x} = \frac{2}{\omega^2-4}$

$\frac{\omega^2-4}{2} = \frac{2}{\omega^2-4} \Rightarrow$

(9) $(\omega^2-4)^2 - 2^2 = 0$

(7) & (8) $\Rightarrow$

$$\begin{pmatrix} 4-\omega^2 & 2 \\ 2 & (4-\omega^2) \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$\begin{vmatrix} 4-\omega^2 & 2 \\ 2 & 4-\omega^2 \end{vmatrix} = 0$$

(9) $\Rightarrow$ $\omega^2-4 = \pm\sqrt{2^2} = \pm 2$

$\omega^2 = 4 \pm 2$

$\omega_{1,2}^2 = 2, 6$

$\omega_1 = \sqrt{2}$, $\omega_2 = \sqrt{6}$

$\frac{y}{x} = \frac{B\cos(\omega t+\varphi)}{A\cos(\omega t+\varphi)} = \frac{B}{A}$

$\left(\frac{y}{x}\right)_{mode\,1} = \left(\frac{B}{A}\right)_{mode\,1} \equiv \frac{B_1}{A_1} = \left(\frac{\omega^2-4}{2}\right)_{mode\,1}$

$\Rightarrow \frac{B_1}{A_1} = \frac{\omega_1^2-4}{2} = \frac{2-4}{2} = -1$

$B_1 = -A_1$ (for mode 1, $\omega_1^2 = 2$)

$\left(\frac{y}{x}\right)_{mode\,2} = \left(\frac{B}{A}\right)_{mode} = \frac{B_2}{A_2} = \left(\frac{\omega_2^2-4}{2}\right)_{mode\,2} = \frac{\omega_2^2-4}{2}$.

$= \frac{6-4}{2} = 1$

$B_2/A_2 = 1 \Rightarrow B_2 = A_2$

mode 1: $X_1 = A_1\cos(\omega_1 t+\varphi_1)$, $Y_1 = -A_1\cos(\omega_1 t+\varphi_1)$

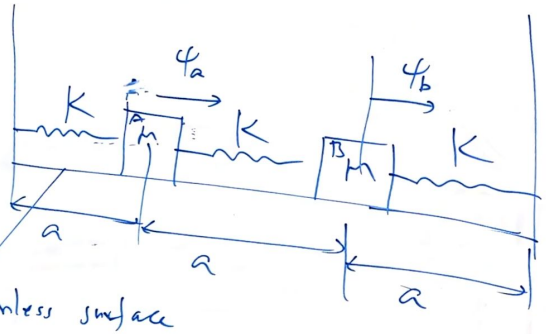
mode 2: $X_2 = A_2\cos(\omega_2 t+\varphi_2)$, $Y_2 = A_2\cos(\omega_2 t+\varphi_2)$

General $X = A_1\cos(\omega_1 t+\varphi_1) + A_2\cos(\omega_2 t+\varphi_2)$

$Y = -A_1\cos(\omega_1 t+\varphi_1) + A_2\cos(\omega_2 t+\varphi_2)$

27.7.22

Waves Lec 21?



A: $M \frac{d^2\psi_a}{dt^2} = -K\psi_a + K(\psi_b - \psi_a)$

B: $M \frac{d^2\psi_b}{dt^2} = -K\psi_b - K(\psi_b - \psi_a)$

Symmetry analysis

Mode 1: $\psi_b = \psi_a \Rightarrow B_1 = A_1$

$$\ddot{\psi}_a = -\frac{K}{M}\psi_a = -\omega_1^2\psi_a$$

$$\Rightarrow \boxed{\omega_1^2 = K/M}$$

$$(\psi_a)_1 = A_1 \cos(\omega_1 t + \varphi_1)$$

$$(\psi_b)_1 = A_1 \cos(\omega_1 t + \varphi_1)$$

Mode 2: $\psi_b = -\psi_a \Rightarrow B_2 = -A_2$

$$M\ddot{\psi}_a = -K\psi_a + K(-\psi_a - \psi_a)$$

$$\ddot{\psi}_a = -\frac{3K}{M}\psi_a = -\omega_2^2\psi_a$$

$$\Rightarrow \omega_2^2 = \frac{3K}{M}$$

$$(\psi_a)_2 = A_2 \cos(\omega_2 t + \varphi_2)$$

$$(\psi_b)_2 = -A_2 \cos(\omega_2 t + \varphi_2)$$

(1)
(2)
(3) ×
(4) ×
(5) –
(6) –
(7)
(8)
(7) =
(8) =